

- Hu, X., Chen, P.-Y., and Ho, T.-Y. Gradient cuff: Detecting jailbreak attacks on large language models by exploring refusal loss landscapes. In *Neural Information Processing Systems*, 2024.
- Hu, X., Chen, P.-Y., and Ho, T.-Y. Token highlighter: Inspecting and mitigating jailbreak prompts for large language models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- Inan, H., Upasani, K., Chi, J., Rungta, R., Iyer, K., Mao, Y., Tontchev, M., Hu, Q., Fuller, B., Testugine, D., et al. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.
- Jain, N., Schwarzschild, A., Wen, Y., Somepalli, G., Kirchenbauer, J., Chiang, P.-y., Goldblum, M., Saha, A., Geiping, J., and Goldstein, T. Baseline defenses for adversarial attacks against aligned language models. *arXiv preprint arXiv:2309.00614*, 2023.
- Ji, J., Hou, B., Robey, A., Pappas, G. J., Hassani, H., Zhang, Y., Wong, E., and Chang, S. Defending large language models against jailbreak attacks via semantic smoothing. *arXiv preprint arXiv:2402.16192*, 2024.
- Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., and Tang, P. T. P. On large-batch training for deep learning: Generalization gap and sharp minima. *International Conference on Learning Representations*, 2017.
- Kumar, A., Agarwal, C., Srinivas, S., Li, A. J., Feizi, S., and Lakkaraju, H. Certifying LLM safety against adversarial prompting. In *Conference on Language Modeling*, 2024.
- Li, H., Xu, Z., Taylor, G., Studer, C., and Goldstein, T. Visualizing the loss landscape of neural nets. *Advances in Neural Information Processing Systems*, 31, 2018.
- Li, X., Zhang, T., Dubois, Y., Taori, R., Gulrajani, I., Guestrin, C., Liang, P., and Hashimoto, T. B. AlpacaEval: An automatic evaluator of instruction-following models. https://github.com/tatsu-lab/alpaca_eval, 5 2023.
- Liu, S., Chen, P.-Y., Kailkhura, B., Zhang, G., Hero, A., and Varshney, P. K. A primer on zeroth-order optimization in signal processing and machine learning. *IEEE Signal Processing Magazine*, 2020.
- Liu, X., Xu, N., Chen, M., and Xiao, C. AutoDAN: Generating stealthy jailbreak prompts on aligned large language models. In *International Conference on Learning Representations*, 2024.
- Ma, X., Gao, Y., Wang, Y., Wang, R., Wang, X., Sun, Y., Ding, Y., Xu, H., Chen, Y., Zhao, Y., et al. Safety at scale: A comprehensive survey of large model safety. *arXiv preprint arXiv:2502.05206*, 2025.
- Mehrotra, A., Zampetakis, M., Kassianik, P., Nelson, B., Anderson, H., Singer, Y., and Karbasi, A. Tree of attacks: Jailbreaking black-box llms automatically. *Advances in Neural Information Processing Systems*, 37:61065–61105, 2024.
- Mitchell, E., Lee, Y., Khazatsky, A., Manning, C. D., and Finn, C. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *International Conference on Machine Learning*, pp. 24950–24962. PMLR, 2023.
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- Peng, S., Chen, P.-Y., Hull, M., and Chau, D. H. Navigating the safety landscape: Measuring risks in finetuning large language models. *Advances in Neural Information Processing Systems*, 2024.
- Qi, X., Huang, Y., Zeng, Y., DeBenedetti, E., Geiping, J., He, L., Huang, K., Madhushani, U., Schwag, V., Shi, W., et al. AI risk management should incorporate both safety and security. *arXiv preprint arXiv:2405.19524*, 2024.
- Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024.
- Ricker, J., Lukovnikov, D., and Fischer, A. AEROBLADE: Training-free detection of latent diffusion images using autoencoder reconstruction error. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9130–9140, 2024.
- Robey, A., Wong, E., Hassani, H., and Pappas, G. J. Smooth-LLM: Defending large language models against jailbreaking attacks. *arXiv preprint arXiv:2310.03684*, 2023.
- Sadasivan, V. S., Kumar, A., Balasubramanian, S., Wang, W., and Feizi, S. Can AI-generated text be reliably detected? stress testing AI text detectors under various attacks. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856.

- Sangwan, R. S., Badr, Y., and Srinivasan, S. M. Cybersecurity for AI systems: A survey. *Journal of Cybersecurity and Privacy*, 3(2):166–190, 2023.
- Tang, Y.-C., Chen, P.-Y., and Ho, T.-Y. Defining and evaluating physical safety for large language models. *arXiv preprint arXiv:2411.02317*, 2024.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Tsai, C.-T., Ko, C.-Y., Chung, I., Wang, Y.-C. F., Chen, P.-Y., et al. Understanding and improving training-free ai-generated image detections with vision foundation models. *arXiv preprint arXiv:2411.19117*, 2024.
- Ullrich, L., Buchholz, M., Dietmayer, K., and Graichen, K. AI safety assurance for automated vehicles: A survey on research, standardization, regulation. *IEEE Transactions on Intelligent Vehicles*, 2024.
- Vaswani, N., Bouwmans, T., Javed, S., and Narayanamurthy, P. Robust subspace learning: Robust pca, robust subspace tracking, and robust subspace recovery. *IEEE signal processing magazine*, 35(4):32–55, 2018.
- Verdoliva, L. Media forensics and deepfakes: an overview. *IEEE journal of selected topics in signal processing*, 14(5):910–932, 2020.
- Xie, Y., Yi, J., Shao, J., Curl, J., Lyu, L., Chen, Q., Xie, X., and Wu, F. Defending chatgpt against jailbreak attack via self-reminders. *Nature Machine Intelligence*, 5(12):1486–1496, 2023.
- Xiong, C., Qi, X., Chen, P.-Y., and Ho, T.-Y. Defensive prompt patch: A robust and interpretable defense of llms against jailbreak attacks. *arXiv preprint arXiv:2405.20099*, 2024.
- Yan, Z., Yao, T., Chen, S., Zhao, Y., Fu, X., Zhu, J., Luo, D., Wang, C., Ding, S., Wu, Y., and Yuan, L. DF40: Toward next-generation deepfake detection. In *Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.
- Ye, J., Wang, Y., Huang, Y., Chen, D., Zhang, Q., Moniz, N., Gao, T., Geyer, W., Huang, C., Chen, P.-Y., et al. Justice or prejudice? quantifying biases in LLM-as-a-judge. *International Conference on Learning Representations*, 2025.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36, 2024.
- Zou, A., Wang, Z., Carlini, N., Nasr, M., Kolter, J. Z., and Fredrikson, M. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.